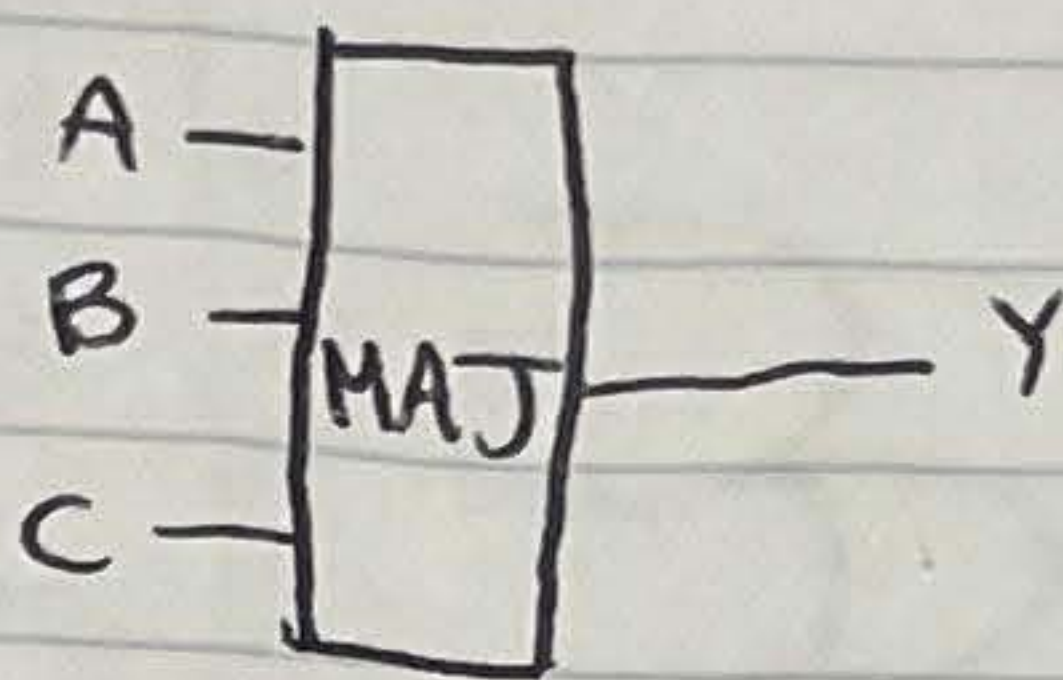


Owen
Manley
11/10/22

Assignment #8 CSC 115

1.) A majority gate produces a TRUE output if & only if more than half of its inputs are TRUE. Complete a truth table for the three-input majority gate shown in Figure 1.

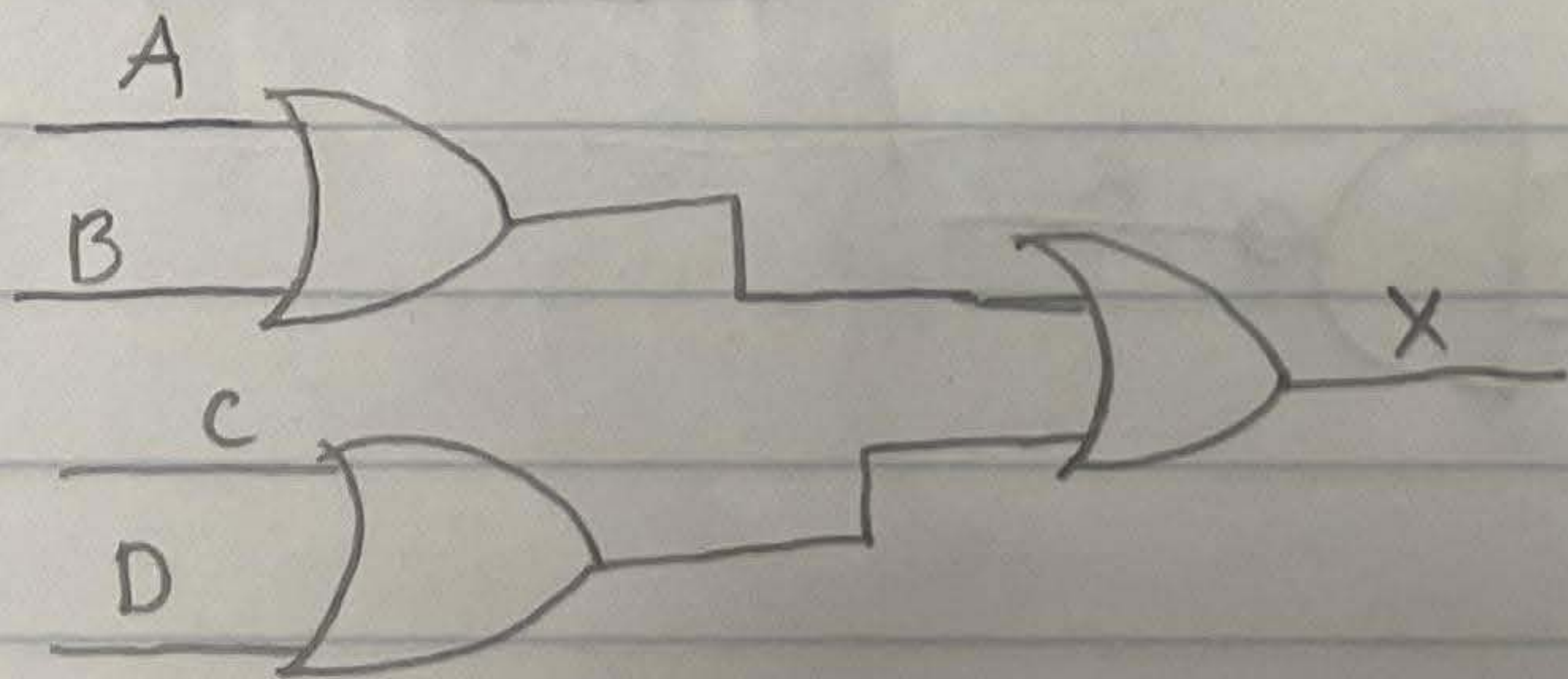


A	B	C	Y
0	0	0	0
1	0	0	0
0	1	0	0
0	0	1	0
1	1	0	1
1	0	1	1
0	1	1	1
1	1	1	1

2.) Draw the symbol, Boolean equation, & truth table

2a.) Four-input OR Gate

Truth Table



Logic Diagram

A	B	(A+B)	C	D	(C+D)
0	0	0	0	0	0
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	1

Boolean Equation

$$X = (A+B+C+D)$$